Fall 2025 Econ H191 paper: Measuring the Impact of Israel's 2021–2023 Port Reforms on Labor Productivity Mediated by Capital Deepening

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Abstract

This paper evaluates the impact of Israel's 2021–2023 port reforms on the sector's persistently low labor productivity and examines the role of capital deepening in mediating these effects. The reforms were aimed at breaking monopoly conditions by introducing intra-port competition through new deep-water terminals in Haifa and Ashdod, and by privatizing Haifa's legacy operator. Besides the overall effect of the reforms on labor productivity, the study quantifies the share of productivity gains associated with capital deepening, a channel of special interest given the uniquely (among comparable OECD members) low levels of capital per worker in Israel's non-tradable sectors. I use event-study difference-in-differences dynamic effect estimators for staggered adoption to evaluate the impact of each of the three reforms on output per worker and the capital-to-labor ratio at the port and terminal-month levels. I then estimate the elasticity of labor productivity with respect to capital–labor ratios and combine these estimates with the event-study coefficients to decompose the reform-induced productivity gains into a component attributed to capital deepening and a residual component.

1 Introduction

Labor productivity in Israel has persistently lagged behind that of comparable advanced economies. Official estimates put GDP per hour worked roughly a quarter below the OECD average, with little sign of convergence over the past three decades.[1] Existing evidence points to accumulated factor gaps – especially a low capital–to–labor ratio – rather than weak total factor productivity as the main driver of this shortfall. Recent development–accounting work finds that physical capital per worker in Israel is around half that of an OECD comparison group largly due to sectors intensive in long–lived structures and equipment [2]. This persistent productivity gap is widely understood to be a major factor behind Israel's high cost of living and poses a central challenge for raising living standards, especially in the country's geographic periphery.

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[1] See, for example, Bank of Israel (2019)

[2] human capital gaps are non-negligible but much smaller than capital per worker gaps

average, with little sign of convergence over the past three decades.[3] Existing evidence points to accumulated factor gaps – especially a low capital-to-labor ratio – rather than weak total factor productivity as the main driver of this shortfall. Recent development-accounting work finds that physical capital per worker in Israel is around half that of an OECD comparison group and that, once both schooling and measured skills are taken into account, differences in accumulated factors explain more than three-quarters of the output-per-worker gap, with low physical capital doing most of the work.[4] This persistent productivity gap is widely understood to be a major factor behind Israel's high cost of living and a key force keeping large segments of low–wage workers and peripheral households in relative poverty despite high aggregate income.

This aggregate shortfall manifests very unevenly across the economy and contributes to domestic inequality. A relatively small, highly productive high-tech sector accounts for a disproportionate share of GDP, exports, and income-tax revenue, while most workers are employed in low-productivity, low-wage non-tradable industries such as retail, construction, transportation, and local services. This dual structure pushes up domestic prices and amplifies inequality: earnings in the high-tech sector have risen with productivity growth, while wages in many non-tradable sectors have largely stagnated. Because these sectoral divides map onto existing ethnic, religious, and geographic cleavages, they reinforce pre-existing inequality and social fractures. These tensions came to the surface in the 2011 social–protest movement over housing costs and living standards, after which the government appointed the Trajtenberg Committee on Socioeconomic Change to examine the cost of living and competition in key non-tradable sectors, including transport and logistics.

Israel's seaports, as the Trajtenberg Committee reported, are a central part of this story. Almost all of Israel's merchandise trade by weight moves through its seaports, and foreign trade amounts to roughly two–thirds of GDP, so the economy is exceptionally dependent on the efficient functioning of a small number of terminals at Haifa and Ashdod. [5] This is especially true given geographic and regional-political factors which leave few realistic alternatives to sea transport. For much of the 2000s and 2010s, Haifa and Ashdod were operated as state–owned regional monopolies that together formed a national duopoly, with powerful unions, rigid work rules, and limited competitive pressure. [move this next part to literature review?] [[[The Trajtenberg Committee found that container output per work team in Israel's ports was 15-25 percent below comparable Middle Eastern ports – and even further below leading global terminals – with resulting efficiency losses estimated in the hundreds of millions of shekels per year. Subsequent audits by the State Comptroller similarly document long ship waiting times, low operational performance relative to international benchmarks, and associated damage to the broader economy, including an increased cost of living. Taken together, these policy reports highlight under–utilized cranes and berths, low labor productivity, and high logistics costs as symptoms of a broader "missing capital" problem in infrastructure–intensive

[3] See, for example, Bank of Israel (2019).

[4] Hazan and Tsur (2019)

[5] Committee for Socioeconomic Change (2011); Israel Ports Company (2007), *Strategic Masterplan for the Development of Israel's Mediterranean Ports*; Essakow (2019); State Comptroller of Israel (2024); and Institute for National Security Studies (2024).

non–tradables, and motivate the recommendation to open the ports to intra–port competition by establishing new deep–water terminals at Haifa and Ashdod.]]]

In response, successive governments launched a series of reforms aimed at restructuring the state-owned port sector, primarily by attracting private investment and introducing intra–port competition. The core of these reforms took place between 2021 and 2023. First, a new deep–water container terminal at Haifa, Bayport, began commercial operations in September 2021 under a long–term operating contract with Shanghai International Port Group (SIPG), bringing a large private entrant into direct competition with the Haifa legacy terminal. Second, a new private container terminal at Ashdod, Southport (Hadarom Container Terminal, HCT), opened to traffic in November 2022, creating intra–port competition there as well. Finally, in January 2023 the government sold a controlling stake in Haifa Port Company to a private consortium led by Adani Ports and Gadot, effectively converting the Haifa legacy terminal from a state–owned enterprise into a privately operated concession within a landlord–port framework. Taken as a whole, these reforms moved Israel's ports toward a model in which multiple private terminal operators and a privatized incumbent compete side by side under long–term concessions. Official documents, economic reports, and political speeches emphasized higher labor productivity, better utilization of existing capital, and ultimately lower logistics costs as key goals of the reform package.

However, despite the scale and visibility of these reforms, there is little causal evidence of their impact on productivity, and even less on the mechanisms through which any gains have occurred. International studies of port reforms and privatization typically document sizable improvements in total factor productivity or throughput per unit of input following corporatization, concessioning, or the entry of private terminals, but they do not separately identify the effects of competition and ownership change, neither do they quantify how much of the productivity improvement operates through capital deepening as opposed to better utilization or organizational change. Israel–specific analyses, in turn, have highlighted long queues, under–utilized cranes and berths, and the potential benefits of competition, but have not used high–frequency micro data or modern difference–in–differences methods to identify the effects of the 2021–2023 reforms on labor productivity or on capital deepening at the terminal level.

However, despite the scale and visibility of these reforms, there is little causal evidence on their impact on productivity, and even less on the mechanisms through which any gains have occurred. International studies of port reforms and privatization typically find sizable improvements in total factor productivity or throughput per unit of input after reforms such as privitizing public port authorities, awarding long–term operating concessions to private terminal operators, or allowing private terminals to enter as competition. Such literature, however, often does not separate the effects of competition from that of privatization, and mostly does not quantify how much of the productivity gains are realized through through capital deepening as opposed to better utilization or organizational change. Israel–specific analyses, in turn, have highlighted long queues, under–utilized cranes and berths, and the potential benefits of competition, but have not used any econometric designs to causally identify the effects of the 2021–2023 reforms on labor productivity or on capital deepening at the terminal or port level.

This paper aims to fill the gaps in the above mentioned literature by attempting a causal evaluation of Israel's recent port reforms on labor productivity and capital-deepening. Using monthly data at the terminal level for Haifa and Ashdod, I construct measures of labor productivity and of the capital–labor ratio for both legacy and entrant container terminals. I then estimate how competition entry and privatization affect these two outcomes using a not–yet–treated event–study difference–in–differences design. For each reform, dynamic effects at each event time are identified by comparing treated terminal–months only to terminal–months that remain untreated (or never treated) with respect to a reform of that kind (competition-introduction or privatization). Finally, I relate the two sets of responses by estimating the elasticity of labor productivity with respect to the capital–labor ratio in a fixed–effects panel regression and using it as the basis for an accounting–style decomposition of the reform effects.

Three questions guide the analysis. First, how did the introduction of private competitors and the privatization of the Haifa legacy terminal affect labor productivity at Israel's container terminals? Second, how did these reforms affect the capital–labor ratio at the same terminals, and do we observe substantial capital deepening in response to competition and privatization? Third, to what extent can the observed changes in labor productivity be accounted for by the observed changes in the capital–labor ratio, given the estimated elasticity of productivity with respect to capital per unit of labor input? By answering these questions, the paper links a macro–level "missing capital" diagnosis to a concrete sector–level reform and asks how much of the post–reform productivity gains can be explained by capital deepening.

TThe paper is organized as follows. Section 2 situations this paper within the broader literature on Israel's productivity gap and on international port reforms; Section **??** describes the data sources and the construction of the key variables; Section 4 sets out the econometric framework; Section 5 presents the empirical results, and Section 6 concludes with policy implications and directions for future research.

2 Literature review

2.1 Israel's productivity gap and capital deepening

A growing macroeconomic literature documents Israel's persistent labor–productivity gap relative to other advanced economies and emphasizes the role of factor accumulation. Brand and Regev (2015a, 2015b) show that productivity gaps between Israel and the OECD average have widened over time, with particularly low productivity in non–tradable sectors, and argue that structural frictions in these sectors are central to the aggregate gap. Hazan and Tsur (2021) conduct a development–accounting exercise and find that most of Israel's output–per–worker shortfall relative to an appropriate European comparison group can be explained by lower physical and human capital per worker. They estimate that physical capital per worker in Israel is roughly one half of the comparison–group level, while human–capital gaps are smaller but still non–negligible. At the industry level, Hazan and Tsur (2021) use a perpetual–inventory method to construct capital stocks by sector and show that Israel's capital gap is especially large in sectors intensive in long–lived structures and equipment, such as construction, utilities, transport, and logistics, whereas high–tech tradables are close to – and in some cases above – peer productivity levels.

Zeira (2021) similarly emphasizes a "missing capital" problem: Israeli output per worker is substantially below that of comparable economies despite relatively high total factor productivity, implying that policies which raise capital intensity in lagging sectors could generate sizable productivity gains. Together, this work points to accumulated factor gaps – especially in physical capital – as a central constraint on Israel's productivity performance, and highlights infrastructure–intensive, non–tradable sectors as natural places to look for bottlenecks.

Ports are a canonical example of such sectors. They are highly capital intensive, dominated by long–lived structures and specialized equipment, and they shape the trade costs faced by the entire economy. Policy documents following the 2011 social protests over the cost of living – especially the Trajtenberg Committee Report (2011) – explicitly identify Israel's seaports as a key non–tradable bottleneck and recommend introducing intra–port competition via new terminals at Haifa and Ashdod. Overall, since Israel's aggregate productivity gap reflects, in part, missing capital in infrastructure–intensive non–tradables, then evidence on how port–sector reforms change capital deepening and labor productivity is directly relevant for the design of macroeconomic policy.

2.2 Port reforms, competition, and productivity

These concerns are not unique to Israel: although Israel's geography and regional politics makes it especially reliant on seaborne trade, many countries face similar bottlenecks due to poorly performing capital–intensive port sectors, and this has motivated a large empirical literature on port reforms and performance. A broad empirical consensus emerges from this literature: intra–port competition – multiple terminals under different ownership within the same port, similar to Israel's recent port reforms – is a key engine of productivity improvement, and capital deepening is an important channel through which competition operates.

Cheon, Dowall, and Song (2010) assemble a global panel of container ports and show that institutional reforms introducing private participation and competition are associated with higher total factor–productivity growth. They stress that productivity gains are more likely when privatization is paired with competitive pressure and technology upgrades, consistent with increased investment in modern handling equipment and yard facilities. Additional studies find that competition is a stronger predictor of productivity gains than privatization alone. Using efficiency–frontier methods and regressions of efficiency scores, Cullinane, Ji, and Wang (2005) and Tongzon and Heng (2005) show that ports with multiple competing terminal operators tend to outperform single–operator ports, including privately owned monopolies, in throughput per unit of input. Most conclusively, The OECD Policy Roundtable on Competition in Ports and Port Services (2011) synthesizes global experience and concludes that introducing intra–port competition generally improves productivity and lowers user costs, whereas privatization without competitive pressure yields more mixed results.

Country–specific case studies are broadly consistent with this pattern. Estache, González, and Trujillo (2002) document efficiency gains in Mexican ports following reforms and decompose them into "frontier shifts" (for example, new investments and technology) and "catch–up" (better utilization of existing capital and labor), though they do not quantify how much of the overall productivity gain is attributable to each mechanism. González and Trujillo (2009) survey port–industry efficiency studies and find that intra–port competition generally yields larger efficiency improvements than privatization alone.

The Israeli port sector has been discussed mainly in policy and think–tank reports rather than in formal econometric work. The Trajtenberg Committee (2011) and subsequent government inquiries locate port inefficiency, driven by monopoly power and rigid labor practices, as an important economy–wide cost and explicitly recommend building new deep–water terminals to induce competition. More recent analyses, such as Ancselovits and Klein (2024), emphasize that historic monopoly conditions at Haifa and Ashdod led to under–utilization of capital and low labor productivity. Official reports and statistical publications document improvements over time in indicators such as ship waiting times, berth occupancy, and throughput, but they do not provide causal estimates of reform impacts or a decomposition of the mechanisms behind observed gains.

2.3 Channels and identification gaps

Overall then, While the port–reform literature frequently mentions multiple mechanisms through which reforms raise productivity – e.g., capital deepening, improved labor utilization, and technical efficiency – these channels are rarely quantified separately. Most studies measure changes in total factor productivity or efficiency scores and interpret them qualitatively, without decomposing how much of the gain is associated with higher capital–to–labor ratios versus better use of existing inputs. As a result, capital deepening is often invoked as an important mechanism, but it remains unquantified.

At the same time, identification in much of this literature is relatively coarse. Empirical work often relies on annual data and standard two–way fixed–effects specifications in staggered–reform settings, in which post–treatment observations for some units can be implicitly used as controls for others. Recent methodological contributions have shown that such designs can yield biased and hard–to–interpret dynamic effects when treatment timing is staggered and treatment effects are heterogeneous over time [cite Sun and Abraham]. In the port context, this raises concerns about how credibly existing studies isolate the impact of individual reforms, especially when competition and privatization are introduced together or compared only across very different ports.

The Israeli case is particularly under–studied along these dimensions. Existing reports document changes in aggregate performance indicators around the 2021–2023 reform wave, but do not provide causal estimates of the separate effects of competition entry and privatization, and no study attempts to distinguish between gains arising from capital deepening and those due to other channels. This paper addresses these gaps by combining high–frequency terminal–level data with modern not–yet–treated event–study estimators and an originally constructed monthly capital series.

It separately traces the dynamic effects of competition entry and privatization on labor productivity and on the capital–labor ratio, and then uses an estimated elasticity of productivity with respect to capital per unit of labor as the basis for a mediation–analysis-style decomposition. This allows me to quantify how much of the observed productivity response is due to (or really, consistent with) capital deepening and how much is attributed to other mechanisms.

3 Data

The core outcomes for this paper are labor productivity (LP) and the capital to labor ration (K/L). To get these outcomes, I make collect throughput, labor, and capital data.

3.1 Monthly throughput, labor input, and productivity

I compile monthly and quarterly throughput series from official publications of the Administration of Shipping and Ports and from port-level statistical yearbooks, published online. The primary quantity measures are containerized throughput in TEU (20-foot-equivalent units) and total cargo in tons. In addition, I collect annual "service indices" reporting TEU per work-hour and various service-quality indicators (e.g., average ship waiting, berth, and stay times) to construct a temporary labor proxy, and to potentially serve in the future as an LP proxy for IV analysis. The table below summarizes the observed throughput series and key performance indicators (KPIs) by port, terminal, frequency, and time range.

Table 1: Observed throughput series and extracted KPIs (coverage and counts)

Series	**Variable**	**Granularity**	**Range**	**N**
Haifa	Throughput (TEU)	Monthly	01-2008–10-2021	166
Haifa	Throughput (tons)	Monthly	01-2008–08-2025	200
Ashdod	Throughput (TEU)	Monthly	01-2008–10-2021	166
Ashdod	Throughput (tons)	Monthly	01-2008–08-2025	200
Haifa–Bayport	Throughput (TEU)	Quarterly	Q3-2021–Q4-2024	14
Haifa–Bayport	Throughput (tons)	Monthly	07-2021–08-2025	50
Ashdod–HCT	Throughput (TEU)	Quarterly	Q3-2021–Q4-2024	14
Ashdod–HCT	Throughput (tons)	Monthly	08-2021–08-2025	49
Haifa	KPI: TEU/hour (Π)	Annual	2018–2024	7
Ashdod	KPI: TEU/hour (Π)	Annual	2018–2024	7
Haifa–Legacy	KPI: TEU/hour (Π)	Annual	2018–2024	7
Ashdod–Legacy	KPI: TEU/hour (Π)	Annual	2018–2024	7
Haifa–Bayport	KPI: TEU/hour (Π)	Annual	2021–2024	4
Ashdod–HCT	KPI: TEU/hour (Π)	Annual	2022–2024	3

While my empirical analysis requires measures of labor input and at the port- and terminal-month level, direct work-hours are not reported at this frequency in public sources. I therefore proceed in two steps. First, I use the annual TEU-per-work-hour KPIs together with observed annual TEU to back out implied annual work-hours for each port and terminal. Second, I construct higher-frequency proxies for labor and that combine these annual KPIs with the observed monthly and quarterly throughput series.

Thus, I calculate LP as:

$$\mathrm{LP}_{p,t} = \frac{\mathrm{TEU}_{p,t}}{H_{p,t}} \quad \text{(port} \times \text{month)}, \qquad \mathrm{LP}_{i,t} = \frac{\mathrm{TEU}_{i,t}}{H_{i,t}} \quad \text{(terminal} \times \text{quarter or month)},$$

where $H_{p,t}$ and $H_{i,t}$ denote actual work-hours in the corresponding period. As the discussion in section 5 shows, this LP proxy is not sufficiently granular, and causes various issues for correct inference. This is, however, only a temporary measure; I am currently in contact with port authorities, and expect to obtain real monthly labor-input data soon. Until such data arrives, I use a temporary LP proxy. The full construction of the labor input series is documented in Appendix **??**.

3.2 Capital data: financial-statement sources

On the capital side, I rely on annual financial statements for the entities that own and operate infrastructure at the Haifa port. These statements provide yearly plant, property, and equipment (PPE) data which I convert into real annual capital stocks. I then convert annual data into monthly series using a simple perpetual-inventory-and-interpolation procedure (Section 3.3; see also Appendix A.3 for implementation details). Table 2 summarizes the resulting PPE stocks and flows for three relevant owners: the legacy operator (Haifa Port Company (HPC)), the landlord and infrastructure owner (Israel Ports Company (IPC)), and the Bayport entrant (Shanghai International Port Group (SIPG)). For each entity and year, the table reports, when such data is available, end-of-year net PPE, gross investment, depreciation, and disposal proceeds. Units are nominal thousands of New Israeli Shekels (kNIS).

Table 2: Annual PPE stocks, infrastructure, and Bayport construction at Haifa by owner (nominal, thousands of NIS)

Panel A: Haifa–Legacy (HPC)

Year	Net PPE (EoY)	Gross inv.	Deprec.	Disposals
2017		25 025	67 739	1 688
2018	1 163 539	30 896	67 012	1 759
2019	1 010 641	25 025	67 739	1 688
2020	987 773	49 285	70 474	497
2021	998 346	75 222	72 280	0
2022	1 006 064			
2023	1 044 165	95 529	68 663	
2024	966 652	61 123	63 266	

Panel B: Haifa–IPC infrastructure

Year	Net PPE (EoY, tot.)	Gross inv. (tot.)	Deprec. (tot.)	Disposals (tot.)
2018	11 516 612	1 501 921	215 476	0
2019	12 513 932	1 100 386	224 640	1 923
2020	12 866 069	560 898	238 299	4 497
2021	13 209 796	419 197	282 698	0
2022	13 143 607	214 152	521 974	13 260
2023	13 105 194	449 072	530 952	22 630
2024	12 829 411	247 552	535 800	5 462

Panel C: Haifa–Bayport (SIPG) CIP

Year	CIP opening (EoY_{t-1})	Additions + cap. int.	Transfers to PPE	Cum. transf. to PPE	CIP closing (EoY)
2018	345	13 528	0	0	13 873
2019	13 162			0	551 635
2020	532 756	538 626	4 482	4 482	1 067 713
2021	1 072 568	663 166	1 339 629	1 344 111	373 212
2022	371 921	133 488	469 804	1 813 915	0
2023				1 813 915	
2024	0	10 518	0	1 813 915	10 518

Panel A (HPC). Net PPE, investment, depreciation and disposals from Haifa Port Company statements; 2017 reports flows but no net stock; the 2022 net stock is reconstructed from the 2023 opening balance; 2023–2024 stocks refer to operating PPE only (excluding investment real estate).

Panel B (IPC). Haifa-related infrastructure PPE aggregated from IPC Note 8; gross investment is Haifa capital expenditure from the IPC cash-flow statement.

Panel C (SIPG). Bayport construction-in-progress balances and flows (additions including capitalised interest, transfers to PPE) converted from RMB to NIS; blank cells indicate items not separately disclosed.

Deflation. All values are nominal thousands of NIS; in constructing real K series they are later deflated using the OECD gross fixed-capital-formation price index for Israel.

A more detailed list of major Haifa projects and Bayport construction-in-progress datasets that complement these financial-statement series is provided in Appendix A.2 (Table 11). Those project-level inputs are used to refine the timing and composition of investment when I construct the monthly capital series.

3.3 From annual accounts to monthly capital series

Starting from the annual PPE stocks and flows in Table 2, I construct real capital series for the Haifa legacy operator (HPC), the Haifa infrastructure owned by IPC, and the Bayport entrant operated by SIPG, and then interpolate them to the monthly frequency used in the panel. Across all three entities, the capital stock is measured in real 2019 NIS, in thousands of NIS, using a common OECD capital-goods price index and three geometric-depreciation scenarios with annual depreciation rates $\delta \in \{0.04, 0.06, 0.08\}$.

For HPC, I treat the PPE series in Panel A as an annual stock–flow system and feed the observed net stocks and flows into the Perpetual Inventory Method (PIM) described in Appendix A.3. The resulting annual data is then disaggregated to months, allocating major projects to their commissioning month and the remainder smoothly within the year, which yields a lumpy monthly operator-capital series $K_{\mathrm{HPC},t}$ for each depreciation rate $\delta \in \{0.04, 0.06, 0.08\}$.

For IPC, I apply the same PIM to the Haifa-specific infrastructure PPE in Panel B, interpreted as landlord civil works shared across terminals. Annual Haifa infrastructure stocks are converted to a smooth monthly series $K_{\mathrm{IPC},t}$ by spreading investment evenly within each year while matching the reported December PPE value.

For SIPG Haifa Bayport, I use the Bayport construction-in-progress (CIP) balances and flows in Panel C, treating transfers from CIP to PPE as annual investment and allowing the capital stock to start only when Bayport becomes operational. Running the same PIM and monthly interpolation on these flows produces a monthly entrant-capital series $K_{\mathrm{SIPG},t}$ consistent with the reported CIP path.

Given these entity-level series, I define a Haifa-cluster capital stock as the sum of the Haifa-K components,

$$K_{\mathrm{Haifa},t} \;=\; K_{\mathrm{HPC},t} + K_{\mathrm{IPC},t} + K_{\mathrm{SIPG},t},$$

computed separately under each depreciation scenario so that the cluster has three alternative monthly capital paths. Next, I combine these monthly capital series with the labor measures described in Appendix A.1 to construct corresponding K/L monthly series at both the terminal level (for Haifa-Legacy and Bayport) and the port level (for the Haifa cluster). These are the final series that enter the event–study and regressions.

3.4 Final LP and K/L series

The throughput, labor, and capital inputs constructed above allow me to create a set of monthly timeseries for LP and K/L. Table 3 summarizes the resulting data.

Table 3: Summary of analysis-ready labor productivity and capital series

Panel A: Labor productivity (LP) series

Series	Level	Freq.	Coverage	Unit
Haifa port LP	Port (Haifa)	Monthly	2018–2021	TEU / work-hour
Ashdod port LP	Port (Ashdod)	Monthly	2018–2021	TEU / work-hour
Haifa–Legacy terminal LP	Terminal (Haifa legacy)	Monthly (from yearly KPI)	2021–2024	TEU / work-hour
Haifa–Bayport (SIPG) LP	Terminal (Haifa Bayport)	Monthly (from yearly KPI)	2021–2024	TEU / work-hour
Ashdod–Legacy terminal LP	Terminal (Ashdod legacy)	Monthly (from yearly KPI)	2021–2024	TEU / work-hour
Ashdod–HCT (entrant) LP	Terminal (Ashdod HCT)	Monthly (from yearly KPI)	2021–2024	TEU / work-hour

Panel B: Capital stocks and implied K/L series (Haifa)

Series	Level	Freq.	Coverage	Unit
Haifa–Legacy operating capital (HPC)	Terminal (Haifa legacy)	Monthly	2018–2024	kNIS, 2019 prices
Haifa infrastructure capital (IPC)	Port infrastructure	Monthly	2018–2024	kNIS, 2019 prices
Haifa–Bayport capital (SIPG)	Terminal (Haifa Bayport)	Monthly	2018–2024†	kNIS, 2019 prices
Haifa cluster capital (HPC + IPC + SIPG)	Port cluster (Haifa)	Monthly	2018–2024	kNIS, 2019 prices

Notes: "kNIS, 2019 prices" denotes thousands of NIS deflated to 2019 using the OECD capital-goods price index. † Bayport capital is zero during the construction phase and becomes positive as assets are transferred from construction-in-progress to PPE; depreciation starts once the terminal becomes operational.

4 Econometric Design

This section sets out the empirical strategy for estimating the effects of the port reforms on labor productivity and capital deepening, and for relating these responses to one another. First, I lay out event–study difference-in-differences (DiD) specifications for $\ln(LP_{i,t})$ (Model 1A) and $\ln(K_{i,t}/L_{i,t})$ (Model 1B), using a not-yet-treated (NYT) design in the spirit of Callaway and Sant'Anna (2) and Sun and Abraham (10). I then present a fixed-effects panel regression of $\ln(LP_{i,t})$ on $\ln(K_{i,t}/L_{i,t})$ (Model 2), which provides an elasticity used in a mediation–style decomposition of the reform effects. Throughout, I discuss the key identifying assumptions and potential threats, and how these are addressed in the empirical implementation.

4.1 Event–study DiD for labor productivity and capital–labor ratios (Model 1)

I treat each reform as an event at the terminal–month level:

- Haifa competition entry: introduction of the private Bayport terminal (09-2021);
- Ashdod competition entry: introduction of the private HCT terminal (11-2022);
- Haifa privatization: sale of the Haifa legacy operator via concession (01-2023).

For each reform r and the terminals it affects, I define an event–time variable $M^r_{i,t}$ measuring the number of months relative to the implementation date of that reform for terminal i: $M^r_{i,t} = 0$ in the month when reform r goes live for terminal i, $M^r_{i,t} = -1, -2, \dots$ in pre–reform months, and $M^r_{i,t} = 1, 2, \dots$ in post–reform months. terminal-months that have not yet or never will undergo a reform of the same kind as reform r (competition entry or privatization) are flagged/enter $M^r_{i,t}$ as controls. The not–yet–treated (NYT) design described below determines which terminal–months are handled as treatment and which as control. For a given reform r, I estimate a pair of event–study regressions with the same specification and sample but different outcomes. For labor productivity,

$$\ln(LP_{i,t}) \;=\; \sum_{m\in\mathcal{M}_r,\, m\neq -1} \beta_m^{LP,r}\,\mathbf{1}\{M_{i,t}^r = m\} \;+\; \gamma_i \;+\; \delta_t \;+\; X_{i,t}'\theta \;+\; \varepsilon_{i,t}^{LP,r}, \tag{1}$$

and for the capital–labor ratio,

$$\ln\!\left(\frac{K_{i,t}}{L_{i,t}}\right) \;=\; \sum_{m\in\mathcal{M}_r,\, m\neq -1} \beta_m^{K/L,r}\,\mathbf{1}\{M_{i,t}^r = m\} \;+\; \gamma_i \;+\; \delta_t \;+\; X_{i,t}'\theta \;+\; \varepsilon_{i,t}^{K/L,r}, \tag{2}$$

where i indexes terminals (or ports in periods before competition entry), t indexes calendar months, γ_i are terminal fixed effects, δ_t are calendar–month fixed effects, and $X_{i,t}$ collects additional controls (e.g. port–specific linear trends or crisis dummies). $\mathcal{M}_r$ denotes the set of months m (leads and lags around reform r) that enter the regression as dummy variables, with the reference month $m = -1$ excluded. That way, each $\beta_m^{LP,r}$ and $\beta_m^{K/L,r}$ measures the change in $\ln(LP)$ or $\ln(K/L)$ at event time m relative to the last pre–reform month, netting out terminal and time fixed effects and controls.

Not-yet-treated design and sample construction. As Sun and Abraham (10) emphasize, in settings with staggered adoption and heterogeneous effects across cohorts and over time, a naive two–way fixed–effects event study implicitly uses already–treated months as controls for later–treated units, which can produce non–transparent and even biased estimates. To circumvent this problem, I implement a *not–yet–treated* (NYT) strategy in the spirit of Callaway and Sant'Anna (2); Sun and Abraham (10). For each reform r and treated terminal, I define treatment at the month when that reform first becomes active for that terminal (event time $M_{i,t}^r = 0$), and then restrict the control months so that treated observations are compared only to terminal–months that have not yet experienced a reform of the same type as r (competition entry or privatization) at month t, or never experience such a reform. More concretely, for each reform r and its corresponding impacted terminals, the sample of (i,t) observations entering (1) and (2) is restricted so that terminal–months with $M_{i,t}^r = m$ (for $m \in \mathcal{M}_r$) are compared only to terminal–months that are not yet treated under a reform of that type at month t; already–treated terminal–months are excluded from the relevant control group.

Table 4 summarizes, for each reform and regression target, the terminal–months used as treatment and as NYT controls. The same NYT windows are used in the $\ln(LP)$ and $\ln(K/L)$ regressions; only the dependent variable differs.

Table 4: Not–yet–treated windows by reform and regression target

Reform	**Regression target**	**Treatment months**	**Control months**
Haifa competition entry (Bayport opens 09-2021)	Haifa-Bayport terminal	Haifa port: 09-2019 to 08-2021 Haifa-Bayport: 09-2021 to 10-2022	Ashdod port: 09-2019 to 07-2021 Ashdod-Legacy: 08-2021 to 10-2022 Ashdod-HCT: 08-2021 to 10-2022
	Haifa-Legacy terminal	Haifa port: 09-2019 to 08-2021 Haifa-Legacy: 09-2021 to 10-2022	Ashdod port: 09-2019 to 07-2021 Ashdod-Legacy: 08-2021 to 10-2022 Ashdod-HCT: 08-2021 to 10-2022
Ashdod competition entry (HCT effective 11-2022)	Ashdod-HCT terminal	Ashdod port: 11-2020 to 07-2021 Ashdod-HCT: 08-2021 to 09-2023	Haifa port: 11-2020 to 08-2021 Haifa-Bayport: 09-2021 to 09-2023 Haifa-Legacy: 09-2021 to 09-2023
	Ashdod-Legacy terminal	Ashdod port: 11-2020 to 07-2021 Ashdod-Legacy: 08-2021 to 09-2023	Haifa port: 11-2020 to 08-2021 Haifa-Bayport: 09-2021 to 09-2023 Haifa-Legacy: 09-2021 to 09-2023
Haifa privatization (Haifa-Legacy sold 01-2023)	Haifa-Legacy terminal	Haifa port: 01-2021 to 08-2021 Haifa-Legacy: 09-2021 to 09-2023	Haifa-Bayport: 09-2021 to 09-2023 Ashdod port: 01-2021 to 07-2021 Ashdod-Legacy: 08-2021 to 09-2023 Ashdod-HCT: 08-2021 to 09-2023

Under the identification assumptions of the NYT design, the event–time coefficients $\beta_m^{LP,r}$ in (1) and $\beta_m^{K/L,r}$ in (2) can be interpreted as causal effects of reform r on $\ln(LP)$ and $\ln(K/L)$ at event time m, relative to the last pre–reform month. The key assumption is that, in the absence of reform r, treated terminals and the corresponding NYT comparison group would have followed similar trends in labor productivity and in the capital–labor ratio over the estimation window. In Section 5, I will assess this assumption using pre–trend patterns and placebo reform dates.

Dynamic effects and window averages. Running the Model 1 regressions yields dynamic treatment–effect estimates for each reform clock. Unlike settings suited to regression–discontinuity designs, where the parameter of interest is the jump exactly at the cutoff, I expect competition entry and privatization to affect labor productivity and the capital–labor ratio gradually over many months after implementation. Focusing on a single event–time coefficient would therefore miss most of the adjustment path. Instead, I treat the post–reform event–study coefficients $\{\beta_m^{LP,r} : m > 0\}$ and $\{\beta_m^{K/L,r} : m > 0\}$ as dynamic paths of the average treatment effect for reform r.

Following the modern DiD event–study literature (e.g. 2; 10), I summarize these paths by taking simple averages over pre–specified event–time windows. For any window $[a, b]$, I define

$$\overline{\beta}_{[a,b]}^{LP,r} = \frac{1}{b-a+1} \sum_{m=a}^{b} \beta_m^{LP,r}, \qquad \overline{\beta}_{[a,b]}^{K/L,r} = \frac{1}{b-a+1} \sum_{m=a}^{b} \beta_m^{K/L,r}. \tag{3}$$

Each window estimate is a linear combination of the underlying event–time coefficients and can be interpreted as the average causal effect of reform r on $\ln(LP_{i,t})$ or $\ln(K_{i,t}/L_{i,t})$ over that window.

In the results tables, for each regression I report five window averages: (i) a “full post” window $m \in [1, M_{\text{post}}]$, where M_{post} is the last observed post–reform month, summarizing the average effect over the entire post period; (ii) a common two–year window $m \in [1, 24]$, which allows direct comparison across ports and reforms; (iii) a first post year $m \in [1, 12]$, capturing short–run effects; (iv) a second post year $m \in [13, 24]$, capturing whether effects persist, strengthen, or decay; and (v) an “average pre” window $m \in [-13, -2]$, which serves as a pre–trend check. Under the identifying assumptions of the NYT design, this average–pre window should be close to zero and statistically insignificant. Window averages are converted to percentage changes in labor productivity and in the capital–labor ratio via $100 \cdot (e^{\overline{\beta}_{[a,b]}} - 1)$

[put the below paragraph in the notes of the table?] All Model 1 regressions are estimated by ordinary least squares on the terminal–month panel described in Table 1, with standard errors clustered at the terminal level.

4.2 Model 2: Elasticity of labor productivity with respect to the capital–labor ratio

Structural equations (1) and (2) from Model 1 characterize how the reforms affect labor productivity and the capital-labor ratio separately. To relate these two sets of responses, I estimate the elasticity of labor productivity with respect to the capital-labor ratio in a fixed–effects panel regression. The baseline specification is

$$\ln(LP_{i,t}) \;=\; \eta \, \ln\!\left(\frac{K_{i,t}}{L_{i,t}}\right) \;+\; \alpha_i \;+\; \tau_t \;+\; u_{i,t}, \tag{4}$$

where η is the elasticity of labor productivity with respect to the capital–labor ratio, α_i are terminal fixed effects, and τ_t are time fixed effects. In a Cobb–Douglas benchmark with constant returns to scale, this elasticity would coincide with the capital share. Here it is a reduced–form elasticity capturing within–terminal co–movements of $\ln(LP_{i,t})$ and $\ln(K_{i,t}/L_{i,t})$ after controlling for time–invariant heterogeneity.

Estimation is based on the subset of terminal–month observations for which both $\ln(LP_{i,t})$ and $\ln(K_{i,t}/L_{i,t})$ are observed and reliably measured [for now it seems it will be both Haifa terminals, 2018-2024]. [leave this next note for the notes section of a robustness table?] As in Model 1, standard errors are clustered at the terminal level, and TEU-weighted versions of (4) will be reported as robustness checks.

Identification in Model 2 relies on the assumption that, conditional on terminal and time fixed effects, variation in the capital-labor ratio is sufficiently exogenous with respect to transitory shocks to labor productivity for the purposes of an accounting–style exercise. Reverse causality (where temporary improvements in productivity induce additional investment) and omitted time–varying shocks remain concerns, which motivate the use of lagged specifications [I have yet to implement this in my code and results tables. Reminder to self to do it], crisis controls in robustness checks, and comparisons of the estimated η to capital–income shares from Haifa Port Company's financial accounts. Throughout, I treat η as an interpretive elasticity rather than a fully identified structural parameter.

Identification in Model 2 relies on the assumption that, conditional on terminal and time fixed effects, within–terminal variation in the capital–labor ratio is not predominantly driven by short–run shocks to labor productivity. Reverse causality (e.g. temporary improvements in productivity inducing additional investment) and omitted time-varying shocks remain potential concerns. Therefore, in the empirical design I [to be determined, I have some ideas] ..... Throughout, I interpret η as a reduced–form elasticity that is useful for the subsequent accounting decomposition, rather than as a fully identified causal parameter.

4.3 Mediation-style decomposition of reform effects

Finally, I combine the event–study estimates from Model 1 with the elasticity estimate $\hat{\eta}$ from Model 2 to construct a mediation–style decomposition of the reform effects on log labor productivity. The goal is to quantify how much of the observed change in $\ln(LP_{i,t})$ is *consistent* with the estimated elasticity and the observed change in $\ln(K_{i,t}/L_{i,t})$, without claiming a fully identified causal mediation parameter.

For each reform r and event–time horizon $[a,b]$, define the total effect of reform r on log labor productivity over that horizon as the window average

$$TE^{r}_{[a,b]} \;=\; \overline{\beta}^{LP,r}_{[a,b]}, \tag{5}$$

and the corresponding average change in ln(K/L) as

$$\Delta C^{r}_{[a,b]} \;=\; \overline{\beta}^{K/L,r}_{[a,b]}. \tag{6}$$

Given the estimated elasticity $\hat{\eta}$ from (4), the component of the productivity response that is consistent with capital deepening over horizon $[a,b]$ is approximated by

$$ME^{r}_{[a,b]} \;\approx\; \hat{\eta}\cdot\Delta C^{r}_{[a,b]}, \tag{7}$$

and the residual component is

$$RE^{r}_{[a,b]} \;=\; TE^{r}_{[a,b]} - ME^{r}_{[a,b]}. \tag{8}$$

And the share of the total effect accounted for by capital deepening is

$$S^{r}_{[a,b]} \;=\; \frac{ME^{r}_{[a,b]}}{TE^{r}_{[a,b]}}. \tag{9}$$

It is important to note that this decomposition is best viewed as an accounting exercise rather than a casual analysis, as it does not rely on an external instrument for $\ln(K_{i,t}/L_{i,t})$ and does not deliver a formal IV–based mediation effect. Instead, it uses the within–terminal elasticity estimated in Model 2 together with the reform–induced changes in $\ln(K_{i,t}/L_{i,t})$ from Model 1 to quantify how much of the event–study response of labor productivity is consistent with capital deepening, and how much remains to be explained by other channels such as improved utilization of existing capital, changes in work organization, or pure efficiency gains.

5 Results and discussion

This section presents the empirical results achieved by running the models described in the previous section. I begin with descriptive evidence on throughput, labor productivity, and capital per worker. Then I report not–yet–treated (NYT) event–study estimates of ln(LP) (Model 1A) and $\ln(K/L)$ (Model 1B), followed by fixed–effects elasticity estimates linking the two outcomes (Model 2). Finally, I combine these estimates in a mediation–style accounting exercise to gauge how much of the reform–induced productivity changes can be reasonably attributed to capital deepening.

Throughout this section I focus on window–average summaries of the event–study paths, deferring full dynamic coefficients to the appendix.

5.1 Descriptive patterns in labor productivity,

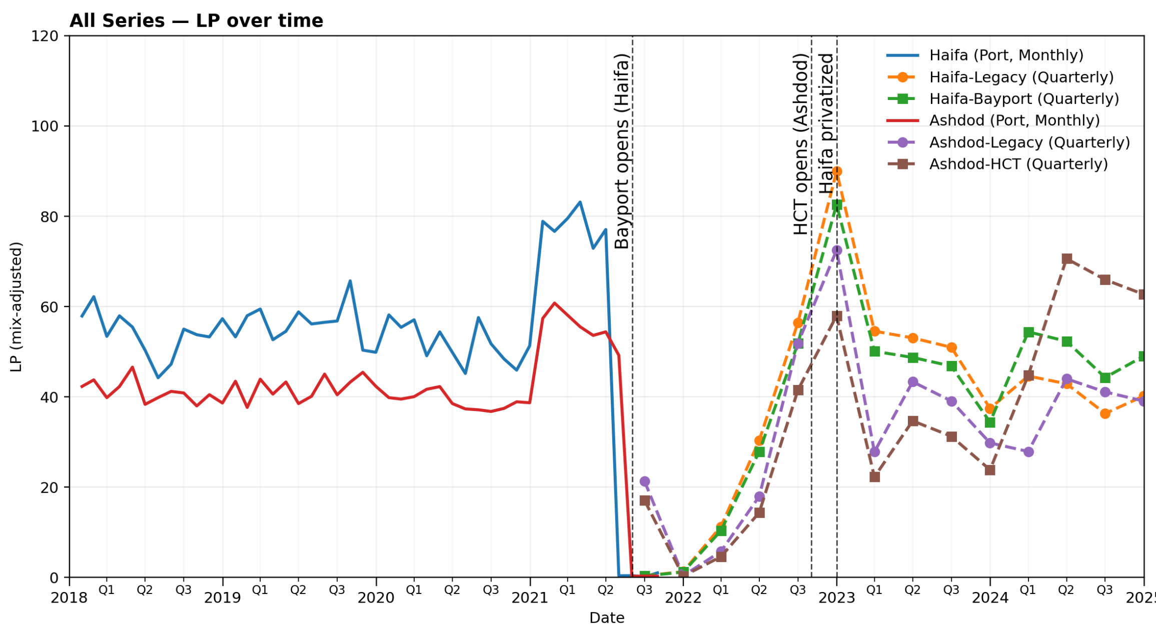

Figure 1: Labor productivity (LP) by port and terminal, 2018–2025. Monthly port-level series are shown for Haifa and Ashdod; quarterly series are shown for entrant and legacy terminals. Vertical dashed lines mark the three reforms: Bayport opening (Haifa, 09–2021), HCT opening (Ashdod, 11–2022), and Haifa legacy privatization (01–2023).

Figure 1 plots labor-productivity series by port and terminal. The most immediate lesson from the figure is methodological rather than substantive: the labor proxy built from annual TEU-per-hour "service indices" is simply not powerful enough to recover month-by-month reform effects. Around each of the three reform dates – Bayport opening at Haifa (09-2021), HCT opening at Ashdod (11-2022), and Haifa-Legacy privatization (01-2023) – the LP paths show long stretches of quarters with values clustered very close to zero, both just before entry (when terminals are structurally assigned zero TEU and zero hours) and in the first few quarters after entry. This pattern is a mechanical by-product of the imperfect labor proxy: the real granularity of the observed labor data is yearly, and the backing out of such data into monthly observations flattens out the data considerably – with especially unreliable results around the event months.

In other words, the yearly KPIs used to recover work-hours are too coarse for the high-frequency variation that a monthly event-study design is trying to detect. The LP paths in Figure 1 should thus be read as year-level productivity signals rather than genuine measures of monthly productivity. Until true terminal-month hours become available from the port authorities, the results based on this proxy are best interpreted as a proof of concept for the econometric design, with any month-specific effect sizes treated with considerable caution.

Despite this limitation, the exercise is still informative as a proof of concept for the econometric design. Figure 1 shows that the constructed LP series are at least broadly sensible in levels and trends (e.g., they rise with the post-entry ramp-up of throughput at entrant terminals and track the broad dynamics at legacy terminals), and they provide just enough variation to implement the not-yet-treated (NYT) event-study framework described in Section **??**. The results that follow should therefore be read primarily as an illustration of the identification strategy and as a lower bound on what more granular labor data could reveal, rather than as precise estimates of the structural effects of reform on productivity.

5.2 Competition entry and labor productivity (Model 1A)

Table 5 reports NYT event–study window averages for $\ln(\mathrm{LP}_{it})$ around competition entry, with Panel A using a Bayport (SIPG) clock for Haifa and Panel B an HCT clock for Ashdod, and with separate entrant/legacy estimates with and without port-specific linear time trends.

The estimates are not suitable for causal interpretation: in both ports the pre–trend F–tests strongly reject flat pre–trends ($p < 0.001$) and the "average pre" windows are far from zero, indicating that the proxy LP series were already drifting well before the reform clocks start.

In Haifa (Panel A), the post–entry window averages for both SIPG and the legacy terminal are large and positive (about 0.37–0.58 log points, roughly 45–80%), whereas in Ashdod (Panel B) the corresponding effects for HCT and the legacy terminal are uniformly negative (around -0.15 log points, a $\approx 14\%$ decline); taken together, these patterns underscore how sensitive the LP results are to the imperfect labor proxy and why true monthly hours are needed before making substantive claims about the impact of competition on productivity. However, overall, we do see a dramatic increase in LP following the reforms, as expected.

Table 5: Average Post-Reform Effects on ln(LP) by Port and Terminal

Panel A: Haifa (competition clock)

	SIPG (entrant)		*Legacy terminal*	
	(1) Baseline	(2) +PortTr	(3) Baseline	(4) +PortTr
Full post (all $m \geq 1$)	0.367*** (0.000)	0.551*** (0.000)	0.372*** (0.000)	0.554*** (0.000)
Post year 1, $m \in [1, 12]$	0.400*** (0.000)	0.584*** (0.000)	0.406*** (0.000)	0.587*** (0.000)
Post year 2, $m \in [13, 24]$	0.367*** (0.000)	0.551*** (0.000)	0.372*** (0.000)	0.554*** (0.000)
Average pre, $m \in [-12, -2]$	2.146*** (0.000)	2.112*** (0.000)	2.146*** (0.000)	2.112*** (0.000)
Pre-trends: F-test p-value	< 0.001	< 0.001	< 0.001	< 0.001
Observations	91	91	91	91
Within R^2	1.000	1.000	1.000	1.000

Panel B: Ashdod (competition clock)

	HCT (entrant)		*Legacy terminal*	
	(5) Baseline	(6) +PortTr	(7) Baseline	(8) +PortTr
Full post (all $m \geq 1$)	−0.148*** (0.000)	−0.148*** (0.000)	−0.148*** (0.000)	−0.148*** (0.000)
Post year 1, $m \in [1, 12]$	−0.148*** (0.000)	−0.148*** (0.000)	−0.148*** (0.000)	−0.148*** (0.000)
Post year 2, $m \in [13, 24]$	−0.148*** (0.000)	−0.148*** (0.000)	−0.148*** (0.000)	−0.148*** (0.000)
Average pre, $m \in [-12, -2]$	−0.371*** (0.000)	−0.371*** (0.000)	−0.371*** (0.000)	−0.371*** (0.000)
Pre-trends: F-test p-value	< 0.001	< 0.001	< 0.001	< 0.001
Observations	95	95	95	95
Within R^2	1.000	1.000	1.000	1.000

Outcome is $\ln(\mathrm{LP}_{it})$ at the terminal×month level. All columns are NYT event–study regressions with terminal and calendar–month fixed effects; columns labeled "+PortTr" additionally include port–specific linear time trends. Event time m is measured in months relative to competition entry, with $m = -1$ omitted; not–yet–treated observations under the relevant competition clock serve as controls. Window averages $\overline{\beta}_{[a,b]}$ are equal–weight means of event–time coefficients over the indicated ranges. "Average pre" reports the mean of β_m over the pre–treatment window $m \in [-12, -2]$. "Post year 1" averages β_m over $m \in [1, 12]$. "Post year 2" is defined as $m \in [13, 24]$; in the current data the event–time support does not yet extend fully to $m = 24$, so this window is only partially populated and, for some specifications, coincides with the full available post period $m \in [1, 13]$. "Full post" averages over all available post–treatment event months $m \geq 1$ for each reform clock. Throughout this and subsequent results tables, effects are reported in log points; a convenient translation into percentage changes is $100 \cdot (e^{\overline{\beta}_{[a,b]}} - 1)$, so that, for example, $\overline{\beta}_{[a,b]} = 0.10$ corresponds to about a 10.5% increase in labor productivity. "Pre-trends: F-test p-value" reports the p-value of a joint test that all lead coefficients $m \leq -2$ are zero, based on the conventional cluster–robust F–test. Cluster–robust standard errors (clustered at the port level) are reported in parentheses, and significance stars (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$) are based on cluster–robust normal p–values from the underlying regressions; given the near-saturated fixed–effects structure (within $R^2 \approx 1$ and extremely small residual variance), these p–values should be interpreted with caution and primarily as descriptive diagnostics.

5.3 Haifa privatization and labor productivity (Model 1A)

Table 6: Haifa Privatization — Dynamic and Window-Average Effects on ln(LP)

	Panel A: Haifa–Legacy (privatization)		Panel B: Haifa–Bayport (placebo)	
	(1) Baseline	(2) +PortTr	(3) Baseline	(4) +PortTr
Full post (all $m \geq 1$)	0.085 (0.060)	0.043 (0.031)	0.085 (0.060)	0.043 (0.031)
Post year 1, $m \in [1, 12]$	0.085 (0.060)	0.043 (0.031)	0.085 (0.060)	0.043 (0.031)
Post year 2, $m \in [13, 24]$	–	–	–	–
Average pre, $m \in [-12, -2]$	0.116 (0.081)	0.162 (0.115)	0.116 (0.081)	0.162 (0.115)
Pre-trends: F-test p-value	0.744	0.694	0.744	0.694
Observations	117	117	117	117
Within R^2	0.947	0.948	0.947	0.948

Notes: Outcome is $\ln(\text{LP}_{it})$ at the terminal×month level. All columns are event-study regressions with terminal and calendar-month fixed effects; columns labeled "+PortTr" additionally include port-specific linear time trends. Event time m is defined relative to the privatization of the Haifa–Legacy terminal (January 2023, $m = 0$); the month $m = -1$ is omitted, and not-yet-treated (NYT) observations under the privatization clock serve as controls. Window-average rows report equal-weight means of the event-time effects β_m over the indicated ranges. "Average pre" uses the pre-treatment window $m \in [-12, -2]$. "Post year 1" averages over $m \in [1, 12]$; in the current estimation sample the post-privatization horizon only extends to $m = 8$, so "Full post" and "Post year 1" currently coincide. "Post year 2" is defined as $m \in [13, 24]$; in the current estimation sample this window has no support, so the corresponding cells are left blank ("–"). "Pre-trends: F-test p-value" reports the p-value of a joint test that all lead coefficients in the pre-window $m \leq -2$ are jointly statistically insignificant, based on the conventional cluster–robust F-test. Panel A reports privatization effects for Haifa Legacy; Panel B applies the same clock as a placebo to Haifa Bayport (SIPG), which is never privatized; in the current data the placebo window averages closely track those for Haifa Legacy, consistent with limited additional terminal-specific response to the privatization shock. Cluster–robust standard errors (clustered at the port level) are reported in parentheses; significance stars (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$) would be based on cluster–robust normal p-values from the underlying regressions, but in this table none of the window-average estimates is statistically significant at conventional levels.

Table 6 applies the NYT event–study design to $\ln(\text{LP}_{it})$ around the January 2023 privatization of Haifa-Legacy, reporting window–average effects for the privatized terminal in Panel A and for Haifa-Bayport under the same clock as a placebo in Panel B.

Here the pre–trend F–tests do not reject flat pre–trends ($p \approx 0.7$), so causal design is more plausable. However, the post–privatization window averages are small (about 0.04–0.09 log points, or 4–9%) and statistically indistinguishable from zero – as well as almost identical for the treated terminal and the placebo. Hence, there is little statistical evidence of a LP increase due to privatization alone.

5.4 Capital deepening and elasticity of ln(LP) with respect to $\ln(K/L)$

Table 7 applies the competition- and privatization-clock event studies to $\ln(K_{it}/L_{it})$, but the baseline NYT specification with terminal and calendar-month fixed effects essentially saturates the small Haifa K/L panel (within $R^2 = 1$ for Haifa–Legacy and for the Bayport series), leaving no residual variance and hence no meaningful standard errors or pre-trend tests; I therefore focus on a "relaxed" specification that keeps unit fixed effects but replaces the full set of monthly dummies with a common linear time trend and collapses the treated-only event-time coefficients into four pre/post windows, which restores enough variation for informative window-average estimates.

Table 7: Haifa: Average post-reform effects on $\ln(K/L)$ (legacy, entrant, and port cluster)

	Haifa–Legacy		Haifa–Bayport (entrant)		Haifa port cluster (central K/L)	
Panel A: Haifa competition entry (competition clock, outcome $\ln(K/L)$)	(1) Baseline	(2) Relaxed + trend	(3) Baseline	(4) Relaxed + trend	(5) Baseline	(6) Relaxed TS
All post $m \in [1, 24]$	0.520	0.531***(0.030)	—	1.534***(0.041)	-0.193***(0.062)	-0.144 (0.123)
Post year 1, $m \in [1, 12]$	0.214	0.214***(0.017)	—	1.565***(0.023)	-0.175***(0.062)	-0.143 (0.088)
Post year 2, $m \in [13, 24]$	0.826	0.848***(0.044)	—	1.504***(0.059)	-0.212***(0.071)	-0.144 (0.164)
Average pre, $m \in [-12, -2]$	-0.056	-0.014 (0.014)	—	-1.973***(0.014)	-0.069 (0.064)	-0.076 (0.059)
Pre-trends: F-test p-value	—	0.148	—	—	—	—
Observations	76	76	—	72	38	38
Within R^2	1.000	0.996	—	0.992	0.344	0.351
Panel B: Haifa privatization (privatization clock, outcome $\ln(K/L)$)	*Haifa–Legacy (treated)*		*Haifa–Bayport (placebo)*		*Haifa port cluster (central K/L)*	
	(7) Baseline	(8) Relaxed + trend	(9) Baseline	(10) Relaxed + trend	(11) Baseline	(12) Relaxed TS
All post $m \in [1, 24]$	-0.574	-0.293***(0.030)	—	0.210***(0.057)	0.193*(0.101)	0.252*(0.144)
Post year 1, $m \in [1, 12]$	-0.471	-0.204***(0.017)	—	0.224***(0.033)	0.117 (0.103)	0.156 (0.123)
Post year 2, $m \in [13, 24]$	-0.687	-0.389***(0.044)	—	0.195**(0.083)	0.269***(0.101)	0.349**(0.170)
Average pre, $m \in [-12, -2]$	-0.696	-0.431***(0.014)	—	0.138***(0.026)	-0.045 (0.104)	-0.053 (0.098)
Pre-trends: F-test p-value	—	0.000	—	—	—	—
Observations	74	74	—	74	37	37
Within R^2	1.000	0.994	—	0.711	0.663	0.669

Outcome is $\ln(K_{it}/L_{it})$. Columns (1) and (7) report the baseline Model 1B specification for Haifa–Legacy with terminal and calendar–month fixed effects; for K/L this is saturated ($R^2 = 1$), so standard errors and pre-trend tests are not defined and coefficients are reported without stars. Columns (2), (4), (8), and (10) use the relaxed panel event–study with unit fixed effects and a linear time trend ("Relaxed + trend"), estimated by OLS with heteroskedasticity–robust standard errors (HC1) in parentheses.
Columns (5) and (11) report univariate time–series window–dummy regressions for the Haifa port capital–labor ratio (HPC + IPC + SIPG, central K scenario) without a time trend ("TS baseline"). Columns (6) and (12) add a linear time trend ("Relaxed TS"). These time–series specifications are estimated by OLS with HC1 standard errors in parentheses; pre-trend F-tests for the time–series models are not reported.
Event time m is measured in months relative to the relevant reform: Bayport opening (competition entry) in Panel A and Haifa–Legacy privatization (January 2023, $m = 0$) in Panel B, with $m = -1$ omitted. Not–yet–treated observations provide controls in the Haifa–Legacy panel regressions. Window averages are equal–weight means of event–time coefficients over the indicated ranges (log points); implied percentage changes follow the formula in the notes to Table 5. Entries "—" denote K/L specifications not yet estimated.

In the competition–entry clock (Panel A), Haifa–Legacy's K/L rises steadily after Bayport opens: the relaxed specification implies an average post–entry increase of about 0.53 log points (column 2, $\approx 70\%$), growing from roughly 0.21 in the first post–year to 0.85 in the second, with no strong evidence of pre–trends (average pre close to zero and $p = 0.15$). Bayport's own K/L jumps by about 1.5 log points, reflecting the construction of a very capital–intensive entrant from essentially zero pre–entry, while the Haifa cluster's central K/L is roughly flat to slightly negative, suggesting that the large terminal–level capital-deepening partly reflects reallocation within a relatively stable overall port–level capital–labor path.

As for the privatization reform (Panel B), Haifa–Legacy's K/L drifts downward in the relaxed specification (column 8, roughly -0.29 log points on average), while Bayport's K/L trends modestly upward under the same timing (column 10, about 0.20 log points), consistent with some rebalancing of capital between the two terminals. At the cluster level, however, the Haifa port K/L shows a moderate post–privatization increase: the time–series estimates in columns 11–12 point to an average post effect of roughly 0.20–0.25 log points (around 20–30%), with the second post–year window reaching 0.27–0.35 log points. Overall then, Table 7 suggests large net capital deepening in Haifa port following competition entry, and modest capital deepening following the sale of the

Table 8: Haifa terminals: elasticity of ln(LP) with respect to $\ln(K/L)$ under alternative depreciation rates

	Haifa–Legacy			Haifa–Bayport			Pooled terminals		
	(1) 4%	(2) 6%	(3) 8%	(4) 4%	(5) 6%	(6) 8%	(7) 4%	(8) 6%	(9) 8%
$\ln(K/L)$ coef.	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)
Observations	N	N	N	N	N	N	N	N	N
R^2	R^2	R^2	R^2	R^2	R^2	R^2	R^2	R^2	R^2

Notes: Entries report coefficients from panel regressions of $\ln(\mathrm{LP}_{it})$ on $\ln(K/L)_{it}$ for Haifa terminals i under depreciation rates $\delta \in \{4\%, 6\%, 8\%\}$. Specifications include terminal fixed effects and a linear time trend. Heteroskedasticity-robust (HC1) standard errors are reported in parentheses. Columns (7)–(9) pool both terminals and constrain the elasticity η to be common across units. The pooled central-depreciation estimate in column (8) provides the baseline $\hat{\eta}$ used in Table 9.

legacy terminal.

Elasticity of ln(LP) with respect to $\ln(K/L)$ (Model 2)

Table **??** summarizes a simple time–series regression of ln(LP) on $\ln(K/L)$ for the Haifa port cluster under the central depreciation rate $\delta = 6\%$. The estimated elasticity is large and negative, about -3.97, with a correspondingly large heteroskedasticity–robust standard error of 3.27 and modest explanatory power ($R^2 = 0.29$), so the coefficient is not statistically different from zero at conventional levels and its sign is the opposite of what a standard production–function narrative would suggest. In light of the short sample, the coarse LP proxy, and the fact that the cluster aggregates multiple terminals and cargo types, this estimate is best viewed as a noisy correlation rather than a meaningful structural elasticity.

5.5 Mediation-style decomposition of reform effects

Table 9 is intentionally left as a skeleton at this stage. A mediation-style decomposition that combines $TE^r_{[a,b]}$ from Model 1A, $\Delta C^r_{[a,b]}$ from Model 1B, and the elasticity $\hat{\eta}$ from Table **??** would only be meaningful if both the LP and K/L paths were credibly identified, but the inadaquate Labor-input proxy and the absence of complete capital data for Ashdod legacy and HCT entrant make such a calculation premature.

In the completed version of the project, once I have (i) true terminal–month labor-input data for all ports and terminals and (ii) full monthly capital series for Ashdod legacy and Ashdod–HCT comparable to those already constructed for Haifa, Table 9 will report, for each reform and horizon, the total LP effect $TE^r_{[a,b]}$, the implied change in $\ln(K/L)$, the mediated component $ME^r_{[a,b]} \approx \hat{\eta}\, \Delta C^r_{[a,b]}$, and the residual $RE^r_{[a,b]}$ and share $s^r_{[a,b]}$. This table will be the eventual overall evaluation of the impact of each reform on Israel's port sector's labor productivity, and what part of these improvements were mediated by capital-deepening.

Checking for output truncation in la

Table 9: Mediation-style decomposition of Haifa reform effects on ln(LP)

Horizon $[a,b]$	$TE^r_{[a,b]}$ (total LP)	$\Delta C^r_{[a,b]}$ (K/L)	$ME^r_{[a,b]}$ (mediated)	$RE^r_{[a,b]}$ (residual)	$s^r_{[a,b]}$ (share)
Panel A: Haifa competition entry (Bayport opening)					
$[1,24]$	β (SE)	β (SE)	β (SE)	β (SE)	s
$[1,12]$	β (SE)	β (SE)	β (SE)	β (SE)	s
$[13,24]$	β (SE)	β (SE)	β (SE)	β (SE)	s
Panel B: Haifa privatization (Haifa–Legacy sale)					
$[1,24]$	β (SE)	β (SE)	β (SE)	β (SE)	s
$[1,12]$	β (SE)	β (SE)	β (SE)	β (SE)	s
$[13,24]$	β (SE)	β (SE)	β (SE)	β (SE)	s

$TE^r_{[a,b]}$ is the Haifa *port-cluster* window–average effect of reform r on ln(LP) from Model 1A, constructed by aggregating terminal-level event–study coefficients for Haifa–Legacy and Haifa–Bayport (see text for details). $\Delta C^r_{[a,b]}$ is the corresponding Haifa port-cluster window–average effect on $\ln(K/L)$ from Model 1B (central depreciation rate $\delta = 6\%$), based on the port-cluster capital–labor ratio series. The mediated component is $ME^r_{[a,b]} \approx \hat{\eta}\,\Delta C^r_{[a,b]}$, where $\hat{\eta}$ is the pooled Haifa elasticity of ln(LP) with respect to $\ln(K/L)$ reported in column (8) of Table 8. The residual component is $RE^r_{[a,b]} = TE^r_{[a,b]} - ME^r_{[a,b]}$, and $s^r_{[a,b]} = ME^r_{[a,b]}/TE^r_{[a,b]}$ (when $TE^r_{[a,b]} \neq 0$) is the share of the total LP effect that is quantitatively consistent with capital deepening through K/L. All effects are reported in log points; shares $s^r_{[a,b]}$ are unitless ratios.

Table 10: Mediation-style decomposition of Haifa reform effects on ln(LP): terminal-specific and pooled Haifa

Unit i	Horizon $[a,b]$	$TE^{r,i}_{[a,b]}$ (total LP)	$\Delta C^{r,i}_{[a,b]}$ (K/L)	$ME^{r,i}_{[a,b]}$ (mediated)	$RE^{r,i}_{[a,b]}$ (residual)	$s^{r,i}_{[a,b]}$ (share)
Panel A: Haifa competition entry (Bayport opening)						
Haifa–Legacy (terminal)	$[1,24]$	0.554	-1.543	11.131	-10.577	20.08
		(0.000)	(0.399)	(3.097)	(3.097)	
	$[1,12]$	0.587	-0.938	6.769	-6.182	11.52
		(0.000)	(0.222)	(1.744)	(1.744)	
	$[13,24]$	0.157	-2.147	15.493	-15.336	98.38
		(0.000)	(0.576)	(4.452)	(4.452)	
Haifa–Bayport / SIPG (terminal)	$[1,24]$	0.551	1.534	3.124	-2.573	5.67
		(0.000)	(0.041)	(0.548)	(0.548)	
	$[1,12]$	0.584	1.565	3.186	-2.602	5.45
		(0.000)	(0.023)	(0.554)	(0.554)	
	$[13,24]$	0.154	1.504	3.063	-2.908	19.85
		(0.000)	(0.059)	(0.544)	(0.544)	
Haifa pooled (Legacy + Bayport)	$[1,24]$	0.554	-0.144	-0.117	0.671	-0.21
		(0.000)	(0.123)	(0.122)	(0.122)	
	$[1,12]$	0.587	-0.143	-0.116	0.704	-0.20
		(0.000)	(0.088)	(0.100)	(0.100)	
	$[13,24]$	0.157	-0.144	-0.118	0.275	-0.75
		(0.000)	(0.164)	(0.151)	(0.151)	
Panel B: Haifa privatization (Haifa–Legacy sale)						
Haifa–Legacy (terminal)	$[1,24]$	0.043	-0.345	2.488	-2.445	57.48
		(0.037)	(0.023)	(0.303)	(0.305)	
	$[1,12]$	0.043	-0.234	1.689	-1.646	39.01
		(0.037)	(0.013)	(0.197)	(0.201)	
	$[13,24]$	–	-0.466	3.360	–	–
			(0.033)	(0.419)		
Haifa–Bayport / SIPG (terminal)	$[1,24]$	0.043	0.210	0.428	-0.385	9.88
		(0.037)	(0.057)	(0.138)	(0.143)	
	$[1,12]$	0.043	0.224	0.455	-0.412	10.51
		(0.037)	(0.033)	(0.103)	(0.110)	
	$[13,24]$	–	0.195	0.398	–	–
			(0.083)	(0.183)		
Haifa pooled (Legacy + Bayport)	$[1,24]$	0.043	0.252	0.205	-0.162	4.74
		(0.037)	(0.144)	(0.170)	(0.174)	
	$[1,12]$	0.043	0.156	0.127	-0.083	2.93
		(0.037)	(0.123)	(0.126)	(0.131)	
	$[13,24]$	–	0.349	0.284	–	–
			(0.170)	(0.220)		

Notes: For unit i and reform r, $TE^{r,i}_{[a,b]}$ is the window-average effect on ln(LP) over horizon $[a,b]$ from Model 1A (specification with port-specific linear trends). $\Delta C^{r,i}_{[a,b]}$ is the corresponding window-average effect on $\ln(K/L)$ from Model 1B (central depreciation rate $\delta = 6\%$): terminal rows use the relaxed Haifa-only event-study; pooled Haifa uses the port-cluster time-series specification with a linear trend. $ME^{r,i}_{[a,b]} \approx \hat{\eta}^i \, \Delta C^{r,i}_{[a,b]}$, where $\hat{\eta}^i$ is from Model 2 (terminal rows: TS+trend elasticities; pooled Haifa: pooled-terminals elasticity). $RE^{r,i}_{[a,b]} = TE^{r,i}_{[a,b]} - ME^{r,i}_{[a,b]}$, and $s^{r,i}_{[a,b]} = ME^{r,i}_{[a,b]}/TE^{r,i}_{[a,b]}$ when $TE^{r,i}_{[a,b]} \neq 0$. All effects are in log points; shares are unitless. Dashes indicate insufficient post-reform coverage for that horizon in the underlying LP panel.

6 Conclusion

In the preliminary form of this paper, I was unable to provide causal estimates of the productivity gains in Israel's port sector following the recent 2021-2023 reforms. This is because I do not yet have access to proprietary labor-input data, which I am in the process of receiving. Despite that, the paper in its current form does provide the complete architecture for doing so: the not–yet–treated event–study framework, the construction of monthly K/L paths from financial statements, and the elasticity and mediation analysis. Everything is in place and can be populated mechanically once terminal–month labor series and consistent capital data for Ashdod legacy and HCT become available.

Looking forward, next semester's work will follow two paths. First, I will complete the measurement side of the project by replacing the current proxy-based inputs with measured terminal–month labor series and extending the capital data to Ashdod's legacy and HCT terminals.

The central extension, however, will move beyond the port sector. Once I have credible causal estimates of the reforms' effects on port–level productivity, I plan to study how these shocks propagate upstream and downstream into Israel's non-tradables economy. Conceptually, lower port-related trade costs act as a change in the price and reliability of imported inputs for sectors such as construction, retail, logistics, and local services – all sectors with deep labor productivity gaps compared to their counterparts abroad. Using sectoral data, I aim to trace how the port reforms affect prices, quantities, and productivity in these non-tradable industries, and to assess whether the aggregate welfare gains implied by the port-sector reforms are concentrated in a narrow set of users or spread more widely across the domestic economy. Only after these additional elements are in place—tighter port-sector estimates, a clearer role for capital deepening, and evidence on upstream and downstream spillovers—will it be appropriate to offer policy recommendations about port governance, concession design, and the sequencing of competition and privatization in infrastructure sectors.

7 Works cited

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Appendix A. Data construction

A.1 Labor input, labor proxies, and productivity measurement

The preferred measure of labor input in this paper is the total number of work-hours used in container operations at each port or terminal and month. Ideally, I would observe these hours directly from administrative records. In practice, publicly available sources only report an annual key performance indicator (KPI) for container operations—TEU moved per work-hour—at the port and, more recently, terminal level. This appendix documents how I use the KPI together with throughput data to construct (i) annual labor-hours, (ii) interim higher-frequency labor proxies, and (iii) corresponding measures of labor productivity.

Annual labor hours from TEU-per-hour KPI. Let y index calendar years and i index entities (ports or terminals, depending on the year). Let $\mathrm{TEU}_{i,y}$ denote observed annual container throughput (in TEU) aggregated from the monthly and quarterly data described in the main text, and let $\Pi_{i,y}$ denote the published TEU-per-work-hour KPI for the same entity and year. Under the convention that the KPI reflects the ratio of TEU to work-hours in container operations, the implied annual work-hours are

$$H_{i,y} \;=\; \frac{\mathrm{TEU}_{i,y}}{\Pi_{i,y}}\,.$$

Interpolating labor input to higher frequencies For the dynamic event–study design, annual labor input is not granular enough. One interim way to increase frequency is to scale observed quarterly throughput with the annual KPI. Let $\mathrm{TEU}_{i,y,q}$ denote throughput in quarter $q \in \{1, 2, 3, 4\}$ and year y. A quarterly labor proxy can be defined as

$$L_{i,y,q} \;=\; \frac{\mathrm{TEU}_{i,y,q}}{\Pi_{i,y}}, \qquad q \in \{1, 2, 3, 4\},$$

so that by construction

$$\sum_{q=1}^{4} L_{i,y,q} \;=\; H_{i,y}.$$

This construction implicitly assumes that labor productivity within a year is approximately constant,

$$\Pi_{i,y,q} \;\equiv\; \frac{\mathrm{TEU}_{i,y,q}}{L^{\text{true}}_{i,y,q}} \;\approx\; \Pi_{i,y} \quad \text{for all } q = 1,\ldots,4,$$

where $L^{\text{true}}_{i,y,q}$ denotes the (unobserved) true quarterly work-hours. Given the focus of this study on changes in productivity around the reform dates, I treat this assumption as too strong for the main analysis. I therefore regard $\{L_{i,y,q}\}$ as an interim device and await the receipt of real monthly labor-input data from the port authorities for accurate causal inference.

Labor productivity proxies based on cargo mix and KPI. Until granular hours become available for the full sample, I construct an interim proxy for labor productivity by combining the TEU-per-work-hour KPI with information on the mix of cargo types. The idea is to scale the KPI by a tons-per-TEU factor that reflects the composition of container cargo within total throughput.

Let $\mathrm{tons}_{p,m}$ and $\mathrm{TEU}_{p,m}$ denote, respectively, total tons and TEU handled at port p in month m, and let $\mathrm{KPI}^{\text{TEU/hour}}_{i,y}$ denote the TEU-per-hour KPI for entity i in year y. I first construct a port$\times$quarter KPI as a TEU-weighted average of terminal-year KPIs:

$$\mathrm{KPI}^{\text{TEU/hour}}_{p,q} \;=\; \sum_{i\in p} \alpha_{i,p,q}\, \mathrm{KPI}^{\text{TEU/hour}}_{i,y(q)}, \qquad \alpha_{i,p,q} \;=\; \frac{\mathrm{TEU}_{i,q}}{\sum_{j\in p} \mathrm{TEU}_{j,q}},$$

where $y(q)$ maps quarter q to its calendar year.

Pre-reform (port$\times$month). For port p and month m in the pre-reform period, I define

$$\widehat{\mathrm{LP}}_{p,m} \;=\; \left(\frac{\mathrm{tons}_{p,m}}{\mathrm{TEU}_{p,m}}\right) \mathrm{KPI}^{\text{TEU/hour}}_{p,q(m)},$$

where $q(m)$ maps month m to its quarter. This proxy preserves the identity

$$\widehat{\mathrm{LP}} \;\approx\; \left(\tfrac{\text{tons}}{\text{TEU}}\right) \times \mathrm{KPI}^{\text{TEU/hour}},$$

which I use mainly to align frequencies across the different series.

Post-reform (terminal$\times$quarter; baseline). After competition entry, when terminal-level KPIs become available, I define a terminal$\times$quarter proxy that uses the port-level tons but terminal-level KPI:

$$\widehat{\mathrm{LP}}_{i,q} \;=\; \left(\frac{\sum_{m\in q} \mathrm{tons}_{p(i),m}}{\mathrm{TEU}_{p(i),q}}\right) \mathrm{KPI}^{\text{TEU/hour}}_{i,y(q)},$$

where $p(i)$ denotes the port containing terminal i.

Planned terminal-specific variant. Where terminal-month tons become available, I will replace the port-level mix with a terminal-specific one:

$$\widehat{\mathrm{LP}}_{i,q}^{\mathrm{term}} \;=\; \left(\frac{\sum_{m\in q}\mathrm{tons}_{i,m}}{\mathrm{TEU}_{i,q}}\right)\mathrm{KPI}_{i,y(q)}^{\mathrm{TEU/hour}}.$$

In all cases, these proxies are intended as interim measures that preserve a transparent structure (mix $\times$ KPI) and align frequencies across series. They will be replaced by the direct ratio LP = TEU/hours as soon as consistent monthly or quarterly work-hour data are available.

A.2 Major projects and CIP datasets for Haifa capital series

This subsection summarises the non–financial-statement inputs used to refine the Haifa capital series. For Haifa–Legacy (HPC), a small set of dated, named projects is used to locate lumpy investment in specific terminals and asset classes (berths, cranes, marine fleet, etc.) and to decide whether each item belongs in the operator capital stock relevant for container operations. For Haifa–Bayport (SIPG), construction-in-progress (CIP) and equipment tables digitised from SIPG reports will supply line-item investment flows that enter the Bayport PIM backbone and determine the timing of infrastructure versus superstructure capital. Table 11 lists the projects and planned datasets and indicates which items are incorporated into the project-augmented K series used in the main analysis.

A.3 Capital measurement: from annual PPE to monthly capital stocks

This appendix records the simple Perpetual Inventory Method (PIM) and interpolation procedure used to convert the annual series in Table 2 into real monthly capital stocks for HPC, IPC and SIPG.

Step 1: Deflation and harmonisation. For each entity $e \in \{\mathrm{HPC}, \mathrm{IPC}, \mathrm{SIPG}\}$ and calendar year y, let $N_{e,y}$ denote end-of-year net PPE, $I_{e,y}^{\mathrm{nom}}$ gross investment, $D_{e,y}^{\mathrm{nom}}$ depreciation, and $S_{e,y}^{\mathrm{nom}}$ disposals, all in nominal NIS (or in RMB for SIPG CIP flows). I deflate these series using the OECD gross fixed-capital-formation price index P_y for Israel (normalised to $P_{2019} = 1$), and convert RMB amounts to NIS before deflation:

$$\tilde{N}_{e,y} = \frac{N_{e,y}}{P_y}, \quad \tilde{I}_{e,y} = \frac{I_{e,y}^{\mathrm{nom}}}{P_y}, \quad \tilde{D}_{e,y} = \frac{D_{e,y}^{\mathrm{nom}}}{P_y}, \quad \tilde{S}_{e,y} = \frac{S_{e,y}^{\mathrm{nom}}}{P_y}.$$

For SIPG, $\tilde{I}_{e,y}$ is constructed from Bayport CIP transfers to PPE, rather than from a conventional PPE note.

Table 11: Project- and CIP-level inputs used in Haifa capital series

Item	Cost (kNIS)[a]	First in-service year	Role in K construction
Panel A: Haifa–Legacy (HPC) major projects (Path B)			
Carmel terminal expansion	220 000	2010	Baseline step-up in modern container quay, yard, and crane capacity in the HPC operator capital stock.
Tugboat *Ilan 1*	45 000	2011	Adds a large tug to the marine-fleet component of HPC K, with geometric depreciation from commissioning.
Tugboat *Elad*	43 000	2015	Second major tug investment; expands marine-fleet K and sharpens the timing of fleet capacity growth.
Kishon West rehabilitation	30 000	2020	Lumpy increase in Kishon West berth and yard infrastructure, raising HPC Path B K around 2020.
Kishon West new cranes	100 000	2023	Large crane investment at Kishon West, generating a discrete jump in crane-related K in 2023.
Kishon East reactivation[b]	—	2020	Used primarily as a timing anchor for Kishon East berth and yard rehabilitation in the lumpy K path.
Mobile crane (Kishon)	36 600		Mobile crane and associated equipment; contributes a mid-sized lumpy increase to HPC Path B K.
5 STS + yard gantry modifications	—		Upgrade of ship-to-shore and yard crane capacity; treated as a single lumpy project in the big-projects path.
Sea department transfer to IPC	180 000	2020	Reallocation of marine-fleet and related assets from HPC to IPC; shapes the split between operator and landlord K.
Waterfront relocation	150 000	2024	Large land and infrastructure project; final-period lumpy increase in HPC K tied to new waterfront operations.
Temporary passenger terminal[c]	8 800	2022	Non-core passenger infrastructure; excluded from the container-operations K series used in the main analysis.
Panel B: SIPG Haifa Bayport equipment inputs			
Bayport equipment projects table[d]	—	2018–2024	Project-level STS, yard-equipment, and vehicle purchases; planned use to refine the timing and composition of Bayport superstructure K on top of the CIP-based PIM path.

[a] Nominal values in thousands of NIS as reported in the financial statements or project documentation; in the capital pipeline these are converted to real 2019 kNIS and embedded in project-level PIM paths (see Appendix A.3).
[b] Kishon East reactivation mainly pins down the timing of berth and yard rehabilitation; where costs are not yet fully pinned down, the project is treated as a timing anchor with conservative cost assumptions.
[c] The temporary passenger terminal is treated as non-core to container operations and is therefore excluded from the project-augmented K series used in the empirical analysis.
[d] At present, only the Bayport CIP panel is fully used in the SIPG PIM construction; the equipment projects table will be incorporated in a refined superstructure series once digitisation is complete.

Step 2: Annual PIM backbone. For each depreciation rate $\delta \in \{0.04, 0.06, 0.08\}$, I construct an annual "backbone" capital stock $K_{e,y}(\delta)$ using a standard PIM recursion:

$$K_{e,y}(\delta) = (1-\delta)\,K_{e,y-1}(\delta) + \tilde{I}_{e,y} - \tilde{S}_{e,y},$$

with an initial stock $K_{e,y_0}(\delta)$ chosen so that the recursion is consistent with the earliest reliable observed net stock for that entity.[6] When an observed net stock $\tilde{N}_{e,y}$ is available, I treat it as the target December value and, if necessary, adjust the level of $K_{e,y_0}(\delta)$ so that $K_{e,y}(\delta)$ matches $\tilde{N}_{e,y}$ (up to rounding). Years with flows but no reported stock (e.g. HPC 2017) are handled entirely through the recursion.

Step 3: Monthly interpolation. Let t index months and let $y(t)$ denote the calendar year of month t. For each entity e and depreciation rate δ, I construct a monthly capital stock $K_{e,t}(\delta)$ that:

1. follows a simple geometric decay within the year, and
2. matches the annual PIM backbone at each December: $K_{e,\mathrm{Dec}(y)}(\delta) = K_{e,y}(\delta)$.

Within a given year y, I assume that "background" investment arrives smoothly across months, while project-related investment for large, dated projects (Appendix A.2) is allocated to the commissioning month. Denote monthly investment by $I_{e,t}$ and monthly disposals by $S_{e,t}$, with

$$\sum_{t\in y} I_{e,t} = \tilde{I}_{e,y}, \qquad \sum_{t\in y} S_{e,t} = \tilde{S}_{e,y}.$$

The monthly recursion is then

$$K_{e,t}(\delta) = (1 - \tfrac{\delta}{12})K_{e,t-1}(\delta) + I_{e,t} - S_{e,t},$$

initialised at the start of each year so that the December value coincides with the annual backbone $K_{e,y}(\delta)$. For IPC and SIPG, $I_{e,t}$ is spread evenly within each year (subject to the annual total constraint); for HPC, $I_{e,t}$ is the sum of a smooth "background" component plus lumpy project components in the relevant months.

Step 4: Cluster aggregation and terminal-level K. The resulting entity-level monthly series $K_{\mathrm{HPC},t}(\delta)$, $K_{\mathrm{IPC},t}(\delta)$ and $K_{\mathrm{SIPG},t}(\delta)$ are combined into a Haifa-cluster stock

$$K_{\mathrm{Haifa},t}(\delta) = K_{\mathrm{HPC},t}(\delta) + K_{\mathrm{IPC},t}(\delta) + K_{\mathrm{SIPG},t}(\delta),$$

and then allocated across terminals when constructing terminal-level K/L ratios in the main analysis. The same procedure is repeated for each depreciation rate δ, so that all event-study and production-function specifications can be estimated under alternative capital-depreciation scenarios.

[6] For HPC, the 2022 net PPE stock is reconstructed from the 2023 opening balance; for SIPG, the pre-2021 construction phase has $K_{e,y}(\delta) = 0$ and positive $\tilde{I}_{e,y}$ only through CIP transfers to PPE.

Appendix B. Full Dynamic Event-Time Tables

A.1. Full LP Event-Time Estimates — Haifa

Table 12: Full Event-Time Estimates — Haifa

	SIPG (entrant)			*Legacy*			
m	(1) Baseline	(2) +PortTr	(3) +Tr&Shocks	(4) Baseline	(5) +PortTr	(6) +Tr&Shocks	$N(m)$
-12	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-11	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-10	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-9	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-8	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-7	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-6	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-5	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-4	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-3	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-2	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
0	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
1	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
2	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
3	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
4	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
5	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
6	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
7	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
8	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
9	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
10	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
11	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
12	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
13	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n

Notes: Event time m is measured in months relative to the Haifa competition–entry reform ($m = 0$). The month $m = -1$ is omitted as the reference bin. All specifications include terminal and month fixed effects; "+PortTr" adds port–specific linear trends; "+Tr&Shocks" additionally includes COVID (2020–21) and late–2023 war–shock controls. β_m are event–time coefficients with standard errors in parentheses. $N(m)$ reports the number of terminal×month observations supporting each event–month m under the baseline NYT specification.

A.2. Full LP Event-Time Estimates — Ashdod

Table 13: Full Event-Time Estimates — Ashdod

	HCT (entrant)			*Legacy*			
m	(1) Baseline	(2) +PortTr	(3) +Tr&Shocks	(4) Baseline	(5) +PortTr	(6) +Tr&Shocks	$N(m)$
-4	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-3	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-2	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
0	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
1	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
2	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
3	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
4	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
5	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
6	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
7	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
8	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
9	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
10	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
11	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
12	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
13	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
14	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
15	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
16	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
17	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
18	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
19	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
20	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
21	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
22	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
23	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
24	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n

Notes: Same NYT event-study specification as Table 12, but with event time m defined relative to Ashdod's competition-entry reform. The omitted reference bin is $m = -1$. All specifications include terminal and month fixed effects; "+PortTr" adds port-specific linear trends; "+Tr&Shocks" additionally includes COVID (2020–21) and late-2023 war shock controls. β_m are event-time coefficients with standard errors in parentheses. $N(m)$ reports the number of terminal×month observations supporting each event-month m under the baseline NYT specification.